BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 084 · Lesson 084 · A* Search: g, h and f Scores

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is Dijkstra's tentative distance?

Q2. What does f(n) equal in A*?

Q3. What is an admissible heuristic?

Q4. Binary search target12 in sorted [2,4,6,8,10,12,14]. Show midpoint indexes.

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I Do · Read new knowledge

Key vocabulary

A* — Heuristic-guided shortest-path search.

g(n) — Cost so far from start.

h(n) — Estimated cost to goal.

f(n) — g(n)+h(n) priority.

New learning

Observe — A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

Apply — A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

Evaluate — Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

Worked example

Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.

1. Calculate all frontier f-scores — From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

2. Improve the path to R — Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

3. Confirm the path and heuristic meaning — Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

We Do · Practise with guidance

Find X→Z: X→Y = 4, X→Z = 10, Y→Z = 3. h(X)=7, h(Y)=3, h(Z)=0.

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OPEN starts {X: g=0, h=7, f=7}
After expanding X:
Y: g=?, h=3, f=?
Z: g=?, h=0, f=?
Which candidate is expanded next?
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Compute priorities and choose

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Calculate the two g/f pairs from the edges leaving X. State which candidate A* should expand next and why.

Expand Y

Y has g=4,h=3,f=7 , Z has g=10,h=0,f=10 , so expand Y. Via Y→Z, candidate g is 4+3=7: improve Z from 10 to 7, with predecessor Z←Y.

Finish the trace

Now Z has g=7,h=0,f=7 . Select it as the goal; predecessor chain Z←Y←X gives X→Y→Z , cost 7 . Distinguish the actual path cost g from estimated priority f.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

After expanding A, calculate f for B and C using stated graph/heuristics. [6]

Question 2

After expanding B then C, show improvements to g(C) and g(D); state best path. [6]

Question 3

Compare A* with Dijkstra when h=0 everywhere, and explain risk of overestimation. [6]

Exit Ticket · Answer before checking

A* h(n) is the exact distance to the goal for every possible heuristic.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Known best route cost from source.

Q2. g(n)+h(n).

Q3. Never overestimates true remaining cheapest cost.

Q4. mid3=8 then mid5=12; return index5.

You Do pages · Independent questions

Q1. B g2 [1],h2 [1],f4 [1]; C g5 [1],h1 [1],f6 [1].

Q2. C from5 to3 via B [2], D from6 to4 via C [2], A-B-C-D total4 [2].

Q3. With h0, f=g [2], same priority basis as Dijkstra [2], overestimating h can prioritise unsuitable paths and undermine optimality [2].

Exit Ticket · Model answer

False. It is generally an estimate.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: